

Assignment 4(Error Sheet)

Question 1.

Line No.	Error	Error Type	Error Correction
10	Semicolon missing after int department = 1	Syntax Error	int department = 1;
12	String "90" assigned to int	Type Mismatch	int marks = 90;
25	if(x = 10) uses assignment	Logical Error	if(x == 10)
30	score undeclared	Semantic Error	Declare variable
44	add() undefined	Semantic Error	Define function
52	temp used outside scope	Scope Error	Declare in outer scope
56	arr[10] out of bounds	Runtime Error	Use valid index
60	%d used for float	Format Specifier Error	Use %f
78	result used outside scope	Scope Error	Declare in outer scope
92	Function argument missing	Function Call Error	Pass argument
121	Extra semicolon after while	Logical Error	Remove semicolon
126	char grade = "A"	Type Mismatch	char grade='A'
130	%s used for integer	Format Specifier Error	Use %d
134	NULL pointer dereference	Runtime Error	Assign valid address
138	calculateAverage undefined	Semantic Error	Define function

Question 2.

Line No.	Error	Error Type	Error Correction
9	String to int assignment	Type Mismatch	int marks=90;
15	Missing semicolon after printf	Syntax Error	Add ;
24	total undefined	Semantic Error	Define variable
34	add() prototype missing	Semantic Error	Add prototype
39	Address of register variable	Semantic Error	Do not use &
41	String to int assignment	Type Mismatch	Use integer
48	department outside scope	Scope Error	Declare in outer scope
57	temp outside scope	Scope Error	Declare in outer scope
61	arr[10] out of bounds	Runtime Error	Use valid index
66	Division by zero	Runtime Error	Ensure divisor != 0
70	Assignment in if condition	Logical Error	Use ==
80	Same output in if/else	Logical Error	Fix logic
95	char grade = "A"	Type Mismatch	char grade='A'
115	Uninitialized variable	Runtime Error	Initialize variable
119	NULL pointer dereference	Runtime Error	Assign memory
124	companyCode undefined	Linker Error	Define variable
128	calculateAverage undefined	Semantic Error	Define function
132	String to int assignment	Type Mismatch	Use integer
136	Extra semicolon after while	Logical Error	Remove ;

Question 3.

Line No.	Error	Error Type	Error Correction
21	Missing semicolon	Syntax Error	int choice;
51	Missing colon after default	Syntax Error	default:
72	Boundary check incorrect	Logical Error	if(index>=5)
81	Extra iteration in loop	Logical Error	i<totalStudents
106	Division by zero	Runtime Error	Use totalStudents
111	Float stored in int	Type Mismatch	Use float
113	Float stored in int	Type Mismatch	Use float
134	totalStudent undeclared	Semantic Error	Use totalStudents
136	printGrade undefined	Semantic Error	Define function
138	calculateRank undefined	Semantic Error	Define function
145	String assigned to int	Type Mismatch	Use integer
149	String assigned to float	Type Mismatch	Use float
176	Structure assigned NULL	Type Mismatch	Use delete logic
185	String compared with ==	Logical Error	Use strcmp
195	Float stored in int	Type Mismatch	Use float
212	fprintf in read mode	Runtime Error	Use w/a mode
223	File open not checked	Runtime Error	Check fp!=NULL
225	fscanf on NULL pointer	Runtime Error	Validate fp
164	Index validation missing	Runtime Error	Check range

Question 4.

Line No.	Error	Error Type	Error Correction
74	Capacity check after insertion	Logical Error	Check before insertion
84	Extra iteration	Logical Error	Use <
109	Account number used as index	Logical Error	Search index
129	Withdraw condition incorrect	Logical Error	Fix condition
176	totalAccount undeclared	Semantic Error	Use totalAccounts
178	calculateInterest undefined	Semantic Error	Define function
180	saveAccounts prototype missing	Semantic Warning	Add prototype
188	String assigned to int	Type Mismatch	Use integer
192	String assigned to float	Type Mismatch	Use float
215	Structure assigned NULL	Type Mismatch	Use delete logic
225	String comparison with ==	Logical Error	Use strcmp
236	Float stored in int	Type Mismatch	Use float
236	generateAverage undefined	Semantic Error	Define function
247	fprintf in read mode	Runtime Error	Use w/a mode
258	File open not checked	Runtime Error	Check fp!=NULL
260	fscanf on NULL pointer	Runtime Error	Validate fp
303	Out of bounds access	Runtime Error	Use valid index
307	NULL pointer dereference	Runtime Error	Assign memory
312	Division by zero	Runtime Error	Ensure divisor !=0

Question 5.

21	Missing semicolon	Syntax Error	int choice;
55	Missing colon after default	Syntax Error	default:
79	Capacity check after insertion	Logical Error	Check before insertion
89	Extra iteration in loop	Logical Error	Use <
156	totalBook undeclared	Semantic Error	Use totalBooks
158	saveLibrary prototype missing	Semantic Warning	Add prototype
160	calculateFine undefined	Semantic Error	Define function
168	String assigned to int	Type Mismatch	Use integer
174	String assigned to float	Type Mismatch	Use float
186	String comparison with ==	Logical Error	Use strcmp
199	Structure assigned NULL	Type Mismatch	Use delete logic
220	Float stored in int	Type Mismatch	Use float
220	calculateAveragePrice undefined	Semantic Error	Define function
231	fprintf in read mode	Runtime Error	Use w/a mode
242	File open not checked	Runtime Error	Check fp!=NULL
244	fscanf on NULL pointer	Runtime Error	Validate fp
269	NULL pointer dereference	Runtime Error	Assign memory
273	Out of bounds access	Runtime Error	Use valid index
278	Division by zero	Runtime Error	Ensure divisor !=0
286	Buffer overflow	Runtime Error	Use safe copy
297	Insufficient malloc size	Runtime Error	Use sizeof
300	Write beyond allocated memory	Runtime Error	Stay in bounds
306	Access after free	Runtime Error	Do not access
308	Double free	Runtime Error	Free once